Questions
Lecture – 3

1. State and explain the Biot-Sevart law. What is the vector form of it?
2. What is the value of proportionality constant of the Biot-Sevart law and the value of magnetic permeability of vacuum? Also, the unit of them.
3. Explain the term "The magnetic field inside a conductor which is produced by itself is always zero".
4. Explain the force between two parallel current carrying conductors.
5. Derive an expression for the force between two parallel conductors carrying currents. Hence define ampere.
6. Two long straight conductors are held parallel to each other at a distance 0.3m. If the conductors carry currents 2A and 8A in opposite directions, then what is the distance of the neutral point from the conductor carrying smaller current? Ans: 0.1m
7. Two straight parallel wires carry currents of 200 mA and 1A in opposite directions. If the wires are 20cm apart, then what is the distance of the neutral point from the 1A wire is (in cm)? Ans: 5cm
8. Two concentric circular loops of radii r_1 and r_2 carry clockwise and anticlockwise currents i_1 and i_2. if the centre is a null point, then i_1/i_2? Ans: r_1/r_2
9. Calculate the value of the magnetic field at a distance of 2cm from a very long straight wire carrying a current of 5A. Ans: 2.5×10^{-6}T
10. At a distance of 10 cm from a long straight wire carrying current, the magnetic field is 0.04 T. what will be magnetic field at the distance of 40 cm? Ans: 0.01 T